Today's Agenda :-

Starting 7:05

1. 2 D Matrices Introduction
2. Iterating 2D Matrices
   a) Row Wise
   b) Column Wise
3. Printing Diagonals
4. Row To Column Zero.

Definition

Array :- Collection of same data types.

ex:- int arr[5] = {1, 2, 3, 4, 5}

matrix:-

A 2D array that stores elements in a grid or table format, organised into rows & columns.; acts as an array of arrays.

Declaration

Java :-

name of the ref. variable

No. of columns

```
int [][] mat = new int [N][m];
```

datatype

No. of rows

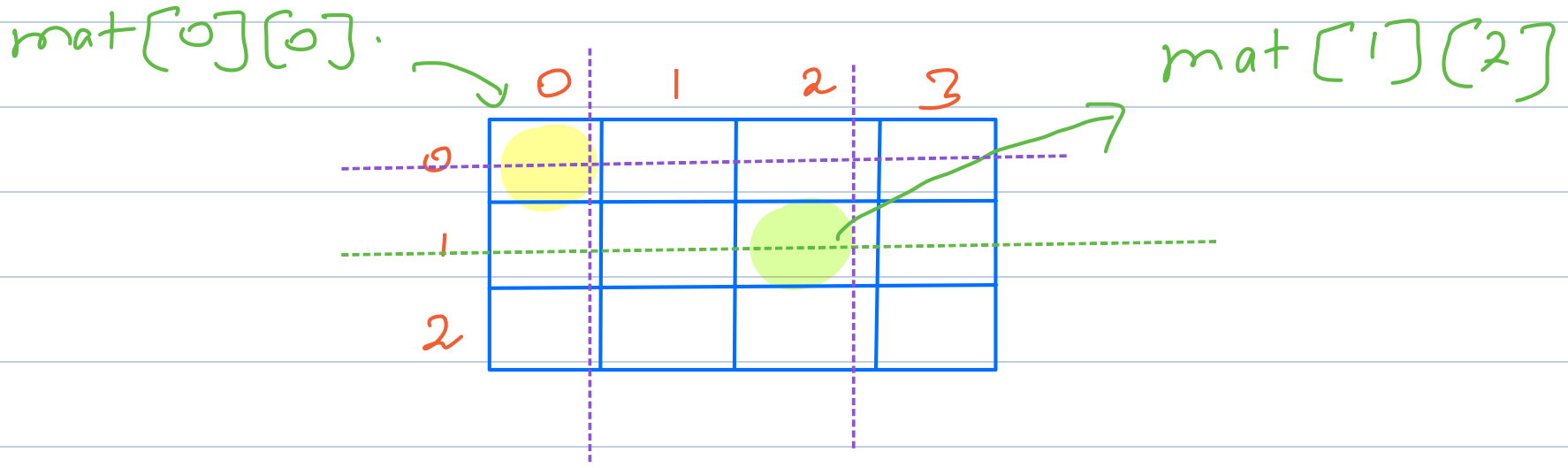
C++ :

```
int mat [N][m];
```

Python :-

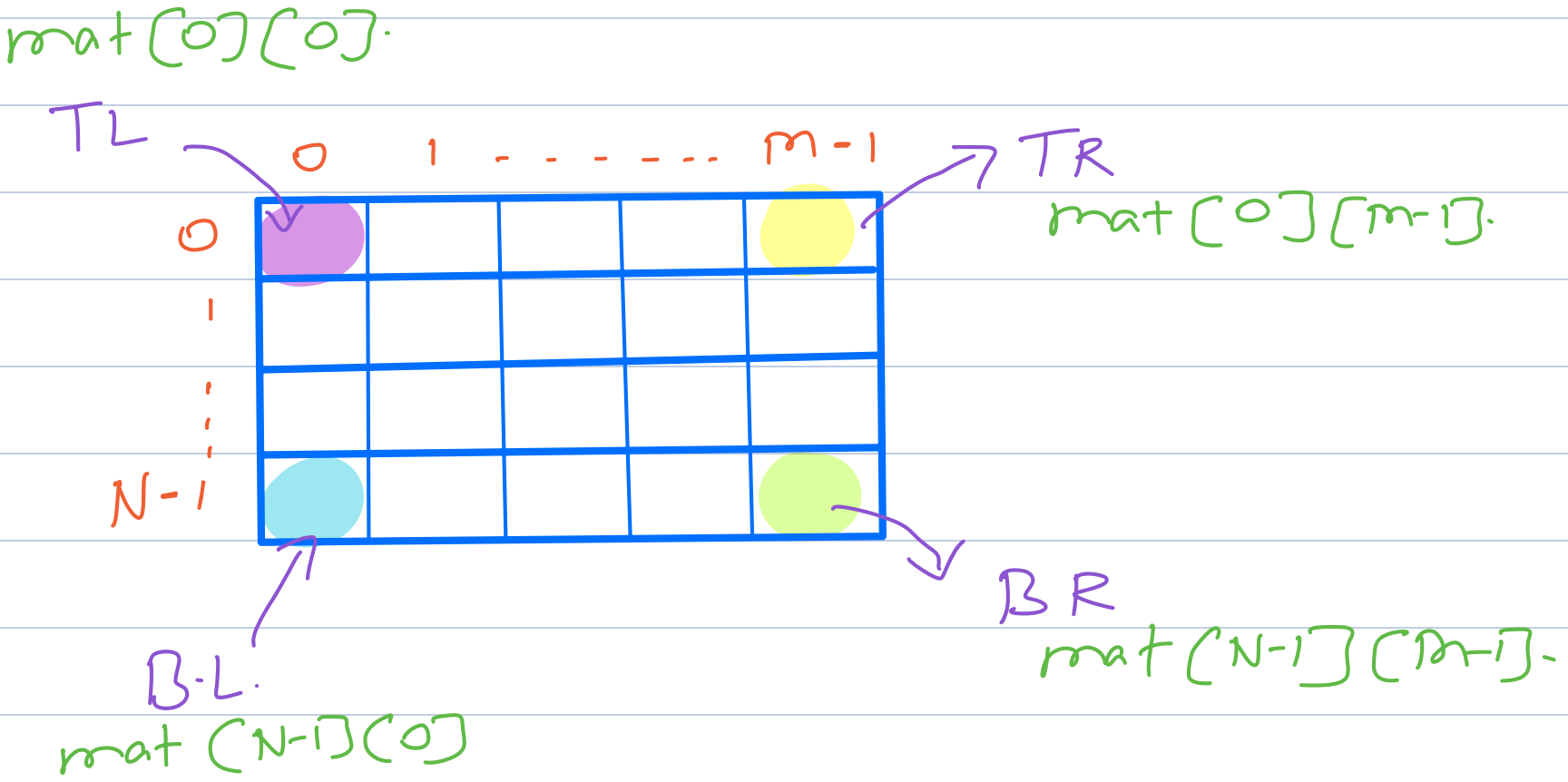
```
mat = [[0] * M for _ in range(N)]
```

Ex:- int mat[3][4]



Quiz 1

Quiz 2



Q:- 1. Given a matrix, print row — wise sum.

ex:- mat [3] [4];

col

row	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12

Output

10
26
42

Sum = 0+1+2+3+4 = 10
= 0+5+6

Approach :

```
void rowSum ( int mat[][]
                , N, m) {

   int sum;

   for ( int row=0 ; row < N; row++){
        sum = 0;
      for (int col=0; col < m; col ++ ){
           sum += mat[row][col];
      }
      print (sum);
   }
}
```

T.C → O(N * m)
S.C → O(1).

Q:-3

Q:- Given a matrix, print col-wise sum.

Ex:-

	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12

Output

15
18
21
24.

```
void colSum ( int mat[][], N, m) {
    int sum;
    for ( int col = 0; col < m; col ++) {
        sum = 0;
        for ( int row = 0; row < N; row ++){
            sum += mat[row][col];
        }
        print (sum);
    }
}
```

T.C → O(N*m)
S.C → O(1).

Q:- Given a square matrix, print its diagonals

N = m

$i + j = N-1$

$j = N-1-i$

	0	1	2
0	1	5	8
1	4	3	1
2	6	5	2

$i == j$

0, 0

1, 1

2, 2.

Principal Diagonal

	0	1	2
0	1	5	8
1	4	3	1
2	6	5	2

0, 2

1, 1

2, 0

Principal Anti Diagonal.

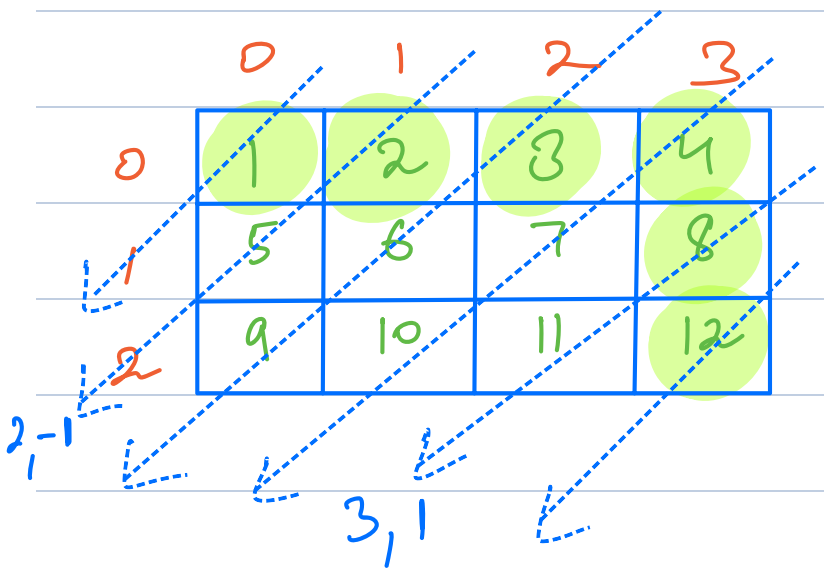
```
void printDiagonals(int mat[][], N){

    # Print Principal Diagonal.
    for(i=0; i<N; i++){
        print(mat[i][i]);
    }
    # Print Principal Anti Diagonal.
    for(i=0; i<N; i++){
        print(mat[i][N-1-i]);
    }
}
```

T.C → O(N) ; S.C → O(1)

Quiz 4

Q:- Print diagonals in a matrix (right to left)

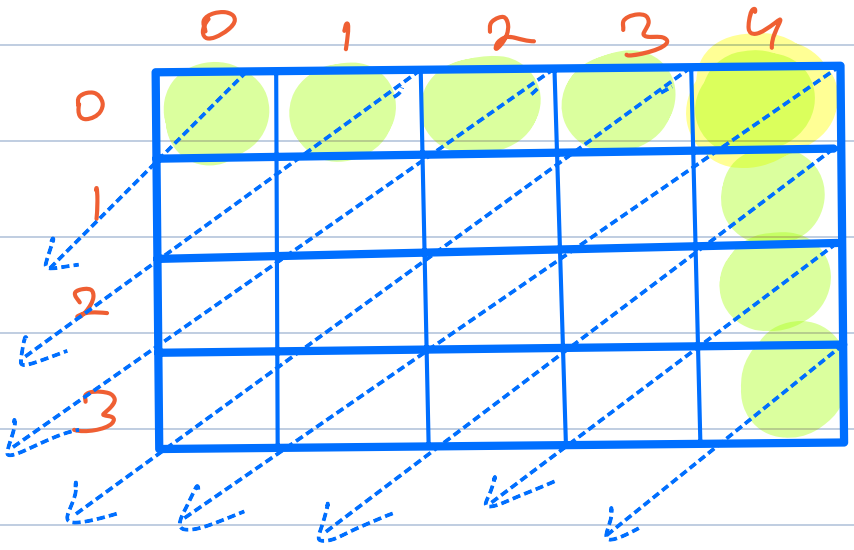



```
row++;
col--;
```

Output:

1
2 5
3 6 9
4 7 10
8 11
12

Quiz 5 $4+5-1 = 8$.



Quiz 6

$M+N-1$

$5+4-1 = 8$.

Observations:-

1. All diagonals movement pattern follow the same row++ & col--;

2. All diagonals either start from the first row or from the last col. making total count as $N+M-1$.

---

1. Fix the S.P. of first row. Move a diagonal until there is a valid cell.

Repeat the process by fixing S.P. in the last col.

i, j → S.P. of a digonal
row, col → For many m that diagonal.

Code

```
void printAllDiagonals(int mat[][], N, m){
    # Fix the S.P of first row.
    i = 0;
    for(int j = 0; j < m; j++){
        row = i; col = j;
        while(row<N && col>=0){
            print(mat[row][col];
            row++; col--;
        }
        println();
    }
    # Fix the S.P of last column.
    j = m-1;
    for(i=1; i<N; i++){
        row = i; col = j;
        while(row<N && col>=0){
            print(mat[row][col];
            row++; col--;
        }
        println();
    }
}
```

8:25.

TC ⇒ O(N*m)
SC ⇒ O(1).

Q:- Given a 2D matrix A.
If a cell A[i][j] == 0, make entire row i as 0 & entire col j as 0.

Ex:-

	0	1	2	3
0	1	2	3	4
1	5	6	7	0
2	9	2	0	4

	0	1	2	3
0	1	2	0	0
1	0	0	0	0
2	0	0	0	0

I/p:

O/p:-

	0	1	2	3	
0	1	2	~~3~~0	~~4~~	0
1	~~5~~0	~~6~~0	7 0	0	
2	~~9~~	~~2~~ 0	0	~~4~~	0
	0				

Ex:- 2:-

	0	1	2	3
0	1	~~2~~0	~~3~~0	4
1	~~5~~ 0	0	~~7~~ 0	~~3~~ 0
2	9	~~2~~ 0	~~2~~	4
			0	

Approach:

In first iteration, simply mark the presence of a zero.

In second iteration, you refer the markers & update the cells.

	0	1	2	3
0	1	2	3	4
1	5	6	7	0
2	9	2	0	4

row :- { 0, 1, 1 3
col :- { 0, 0, 1, 1 3-
0 1 2 3

---

int row[N]; ∀ i row[i] = 0;
int col[m]; ∀ j col[j] = 0;

1st iteration

```
for ( i =0; i <N; i ++) {
    for (j =0; j < m; j ++) {
        if (mat[ i ][j ] == 0) {
            row[ i ] = 1;
            col [j] = 1;
```

3 3 3

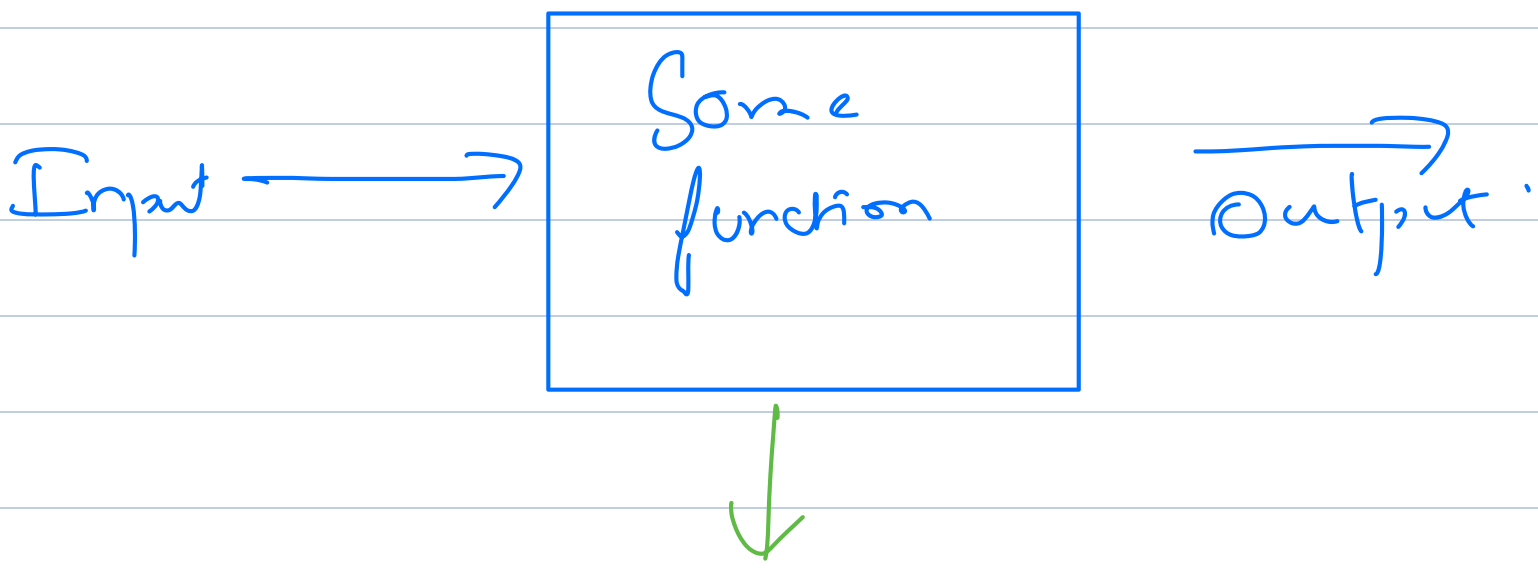
2nd iteration

```
for ( i = 0; i < N; i ++) {
    for (j = 0; j < m; j ++) {
        if (row[i] == 1 || col[j] == 1){
            mat[i][j] = 0;
        }
    }
}
```

T.C → O(N*m)

S.C → O(N+m);

8:45



Any other memory my function used is Sc.